Brain Tumor Detection in MRI Scans: A YOLO-Based Deep Learning Approach

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***Abstract*—Brain tumor detection and classification have become crucial for the successful planning of treatment and better results for patients. Manual interpretation of Magnetic Resonance Imaging (MRI) scans is time-consuming and depends on professional radiologists, causing undesirable diagnostic delays. This study proposes a deep learning approach that uses the YOLO (You Only Look Once) architecture for automatic brain tumor detection and multi-class classification into four classes: Glioma, Meningioma, Pituitary Tumor, and No Tumor Experiments are conducted on the MRI Brain Tumor dataset from Kaggle, comparing several YOLO versions (YOLOv5, YOLOv10, and YOLOv11) and sizes (nano, small, medium). Our findings show that each YOLO version achieved an excellent recall with a specific class; YOLOv5m, YOLOv11m, YOLOv11n, and YOLO10s achieved the best recall for Glioma, Meningioma, Pituitary, and No Tumor classes, respectively. However, the YOLOv5n model had the highest overall recall of 0.902. This study illustrates how YOLO models can greatly improve brain tumor diagnosis, providing an efficient solution for healthcare systems by enabling fast, accurate, and automated MRI interpretation.**

The full implementation code is available at: https://github.com/AlshangitiJumana/BrainTumor.git.

***Index Terms*—Brain Tumor Classification, YOLO, Object Detection, MRI, Computer Vision**

I. Introduction

Brain tumors are a significant health challenge worldwide, Magnetic Resonance Imaging (MRI) is one of the most common nonsurgical and safe methods used to detect brain problems, with early detection and accurate classification being crucial for effective treatment planning. However, manual interpretation of magnetic resonance images is time-consuming and requires substantial expertise, which can lead to diagnostic delays, particularly in regions facing a shortage of specialized medical professionals.

In Saudi Arabia, the healthcare system faces such pressures. A study in nine hospitals reported an average of 20-40 outpatient magnetic resonance imaging performed daily [1]. Despite this demand, only 4,172 radiologists are currently practicing across the Kingdom [2]. The average waiting time for an MRI appointment is approximately 11 days, even in major healthcare centers [3]. Moreover, there are only 238 neurosurgeons in the country, with many regions underserved by specialized surgical care [4]. These statistics highlight the urgent need for intelligent, automated tools that can support and accelerate the diagnostic process. Recent advancements in deep learning and computer vision have introduced powerful models capable of such support. Among them, the "You Only Look Once" (YOLO) family has emerged as a leading architecture for real-time object detection tasks. Initially developed for general-purpose vision problems, YOLO models are increasingly applied in medical imaging due to their ability to process images quickly while maintaining high detection accuracy. YOLO works by scanning an image once and simultaneously predicting bounding boxes and class probabilities, making it particularly suitable for detecting anomalies like tumors in MRI scans [5].

In this research, we propose a YOLO-based approach for both detection and classification of brain tumors using MRI images. Unlike some studies that focus solely on binary detection, our model classifies tumors into four categories: glioma, meningioma, pituitary tumor, and no tumor. This added granularity can significantly enhance clinical decision-making. We base our experiments on the publicly available Labeled MRI Brain Tumor Dataset from Kaggle [6]. This dataset provides a rich collection of labeled MRI images and is widely used for benchmarking classification models. Through comparative evaluation of different YOLO versions, we aim to identify the most effective model for fast, accurate brain tumor diagnosis, helping to bridge the diagnostic gap in overburdened healthcare environments such as Saudi Arabia.

II. Literature Review

Z. Rasheed et al. [7] introduced a new method for brain tumor classification using a convolutional neural network (CNN) and image enhancement techniques. The goal of this study is to use MRI images to classify different types of brain tumors, namely pituitary tumors, meningiomas, gliomas, and tumor-free cases. It utilized a publicly accessible MRI dataset that combines three datasets, including Figshare, SARTAJ, and BR35H, which consists of 7023 grayscale human brain MRI images. The authors enhanced the quality and improved feature extraction of MRI images by incorporating Gaussian-blur-based sharpening and Contrast-Limited Adaptive Histogram Equalization (CLAHE). The proposed study mainly uses a Convolutional Neural Network (CNN) and utilizes pre-trained models such as VGG19, ResNet50, VGG16, MobileNetV2, and InceptionV3 for comparison. The comparison shows that the model outperforms the pre-trained models in performance with accuracy, precision, recall, and F1-score 97.84, 97.85, 97.85, and 97.90%, respectively. However, there are concerns about the applicability of this study to different clinical environments because the model was tested and validated successfully using a specific MRI dataset. Additionally, while image enhancement techniques improve clarity, they may affect classification accuracy in some environments, so they should be tested on different imaging modalities and clinical scenarios to ensure adaptability in real-world applications.

Mathivanan et al. [8] employed deep learning and transfer learning to detect brain tumors. The aim of this study is to enhance the diagnostic accuracy of brain tumor classification. They assessed four transfer learning architectures, including ResNet152, VGG19, DenseNet169, and MobileNetv3. Five-fold cross-validation was employed to fine-tune and train the models. Moreover, they applied image augmentation techniques, such as rotation, flipping, and zooming, to improve the balance of the dataset as well as enhance the models' generalization. They utilized a dataset from Kaggle that consists of 7023 MRI images, which are categorized into four classes: pituitary gland tumors, meningioma, glioma, and normal. The dataset was split into 80% for training and 20% for testing. MobileNetv3, DenseNet169, VGG19 and ResNet152 achieved test accuracies of 98.52%, 97.53%, 95.62% and 96.92%, respectively. One limitation is that the dataset was limited and does not reflect the diversity of real-world brain tumor cases. In addition, the study lacks real-world testing in clinical settings.

In a study by Agarwal et al.[9] aimed to develop a deep learning model for the detection and classification of brain tumors at an early stage. Their study involved two stages. The First one is the image preprocessing stage, which applies a technique called the Optimized Double Threshold Weighted Constrained Histogram Equalization (ODTWCHE) that helps in enhancing contrast and brightness levels of MRI images. Moreover, this step makes the images more visible, enabling better feature extraction. Next, in the second stage, which is the brain tumor detection and classification into either benign or malignant by implementing the Inception V3 model, which is a pre-trained Convolutional Neural Network (CNN) model. In addition, the authors compared the result of this Inception V3 with other CNN models, such as AlexNet, ResNet-50, DenseNet-201, VGG-16, VGG-19, and GoogLeNet. Moreover, this study used the Figshare MRI images dataset, which consists of 3064 images for training and 397 test images, each with different contrast levels. As a result, the Inception V3 model achieved a high accuracy of 98.89%, outperforming the other existing CNN architectures in tumor classification tasks. However, one enhancement suggested for this study is to improve the system computational efficiency to enable real-time clinical applications, hence making it more accessible and practical for different healthcare tasks.

M. F. Almufareh et al. [10] this paper introduce an automated deep-learning framework for brain tumor segmentation and classification using advanced YOLO-based models. The study utilizes two YOLO versions, YOLOv5 and YOLOv7, to detect and classify meningioma, glioma, and pituitary tumors in MRI images. The researchers employed a dataset containing 3064 brain MRI slices. The data image was preprocessed by conversion from .mat to .png format, and the tumor mask annotations were then aligned using custom morphological and polygonal transformations to ensure accurate segmentation. The study used YOLO models because they can achieve real-time object detection while maintaining high accuracy. YOLOv5 achieved precision, recall, and mAP@0.5 values of 93.6%, 90.6%, and 94.7%, respectively, while YOLOv7 achieved comparable results with slightly better generalization over broader IoU thresholds (mAP@0.5–0.95). Compared to traditional object detection approaches such as RCNN, Faster RCNN, and Mask RCNN, the proposed models demonstrated superior accuracy and inference speed. However, the study noted a limitation in glioma detection, which showed high false positives, this limitation may be due to training on a single dataset. So, the approach needs to be expanded to include diverse imaging sources in the future.

The study by Dr. A.D. Katre et al.[11] aims to develop an efficient and cost-effective deep learning-based system for the detection and localization of brain tumors from medical imaging. The aim is to support early and accurate diagnosis, particularly in remote areas where access to advanced medical diagnostics is not readily available. The dataset used consists of 253 T2weighted MRI images, which were sourced from Kaggle and Google and are labeled as tumor and non-tumor classes. These pictures were preprocessed, meaning they were resized to 270×270 pixels and normalized, and data augmentation techniques like rotation, flip, and zoom to enhance model robustness. The strategy is to build a Convolutional Neural Network (CNN) consisting of nine layers (three convolutional, three max pooling, dropout, and two fully connected) and employ a new image processing algorithm called TumoVision for extracting tumor regions. TumoVision employs pseudo-coloring (specifically COLORMAP_JET) and edge detection using the Canny filter to segment out tumor regions based on the removal of specific pixel intensity values. The proposed CNN model achieved a high accuracy of 96.15%, outperforming VGG16's accuracy of 92.31%. In addition, a web interface was established with Flask in order to provide the user a platform to upload or capture MRI scans, which are processed to determine tumor presence and localization. Nevertheless, the work does have some disadvantages, such as a fairly small data set (253 images only) which might hinder generalization, and image quality variability if drawn from multiple sources. Furthermore, hand-crafted color thresholds for tumor extraction might may not perform consistently across all image types or MRI conditions.

III. Gap analysis

Many advancements have been introduced in the field of brain tumor detection; however, prior studies still face several key gaps that limit their applicability in real-world clinical environments. As highlighted in the literature review, one major challenge in brain tumor detection is the limitation of the datasets used for model training. For instance, Dr. A.D. Katre et al. [11] utilized a small dataset that relied on only 253 images, which limited the model's capability to generalize effectively, while Mathivanan et al. [8] trained a model on a limited dataset that did not represent the diversity of brain tumor cases in clinical practice. In addition, some approaches rely on hand-crafted techniques, such as color thresholding for tumor extraction, which may fail to adapt to the variability of MRI scans. Furthermore, the use of image enhancement techniques by Z. Rasheed et al. [7] to improve image clarity may affect classification accuracy. This project uses YOLOv7, YOLOv8, YOLOv9, and YOLOv11 versions of the YOLO object detectors family to fill these gaps. This study aims to detect brain tumors and classify them into four categories: gliomas, meningiomas, pituitary tumors, and no tumors. This study also compares different YOLO versions and model sizes (nano, small, medium) based on precision, recall, parameter size, and inference speed metrics.

IV. Methodology

This study developed an effective YOLO-based system to detect and classify brain tumors in MRI images. It utilized standard YOLO architectures without modifications and then fine-tuned them on the MRI brain tumor dataset to enhance performance. As the focus is on improving the recall metric, which is very important to reduce the probability of missing tumor cases, we evaluated multiple YOLO model versions and sizes, analyzed their performance, and selected the most appropriate model.

A. Dataset description

The labeled MRI Brain Tumor dataset is available on Kaggle [6] for brain tumor classification tasks. It was prepared to support training models like YOLOv7 to identify and localize brain tumors. The dataset includes MRI scans from multiple anatomical planes, including axial, sagittal, and coronal views. The images were annotated by medical experts using the LabelImg tool. The dataset has four classes of clinical diagnosis: 1594 samples of no tumor, 1457 samples of pituitary tumor, 1321 samples of glioma tumor, and 1339 samples of meningioma tumor. All samples provide a comprehensive visual representation of different cases and have been carefully sorted and reviewed by expert radiologists. Figure 1 shows the distribution of MRI images across the four tumor classes.

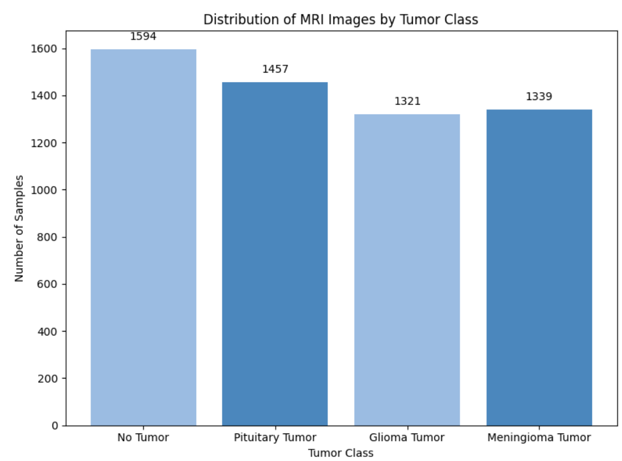

Fig. 1. Distribution of samples in the MRI brain tumor dataset across four diagnostic classes.

B. Baseline

Initially, we utilized the implementation provided by B. Selcuk et al. [12], who trained a YOLOv8 model on the same dataset. We used the same hyperparameters: training for 50 epochs with a batch size of 16.

C. Experimental Setup

This study used YOLO models to classify brain tumors through Python programming language on VS Code. First, the images were resized to meet the model requirements. Then, the dataset was split into 80% for training and 20% for testing. Subsequently, we employed three different YOLO models, including YOLOv5, YOLOv10, and YOLOv11. Each was implemented with three sizes: nano (n), small (s), and medium (m). The training is done using the same hyperparameters: Number of epochs: 50 Batch size: 16. Finally, the models were evaluated using various performance metrics, including Precision, Recall, Number of parameters, and Inference time. Figure 2 below illustrates the steps of the proposed methodology.

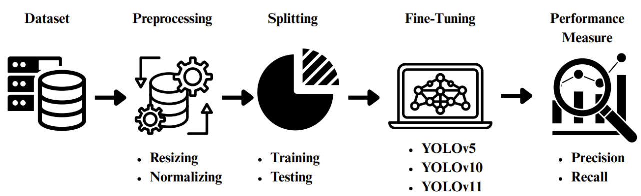

Fig. 2. Proposed Methodology Workflow for Brain Tumor Detection Using YOLO Models

D. Performance Measure

Performance measurement and evaluation are essential steps in determining the model's efficiency. The performance measures that were primarily used in our models are Precision, Recall, Number of parameters, and Inference time. Precision is the percentage of true negative results that are wrongly labeled as positive, indicating the model's accuracy in tumor detection. On the other hand, Recall is the proportion of all true positive results accurately identified as positive, reflecting the model's ability to detect tumors without missing cases. Furthermore, the inference time and the number of parameters were considered to assess model efficiency. Performance metrics for all models are summarized in Table I. A confusion matrix comprising True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN) can be used to implement the performance measurements. Employing different kinds of measures will significantly help when evaluating the performance of the proposed study. Equation (1) illustrates the precision formula, whereas Equation (2) represents the recall formula.

$$\text{Precision} = \frac{TP}{TP + FP} \tag{1}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{2}$$

V. Results and Discussion

This paper, "Brain tumor detection and localization with yolov8" by B. Selcuk et al.[12], used the YOLOv8 model on the same labeled MRI brain tumor dataset from Kaggle, which reported achieving 0.961 recall and 0.976 precision. They shared their implementation code on a GitHub repository, so fortunately, we were able to use their model as our study baseline. We adopted the same hyperparameters the study specified, 50 epochs and 16 batch sizes for training, and acquired the best-preformed model. However, our results differed slightly from theirs; the recall was 0.885, and the precision was 0.913. Based on that, we started our experiment where we tried to increase the recall; our perspective is that the recall in this application forms the most crucial metric since returning all the positive data is more important than minimizing the false positive data that could be easily discarded when running more medical tests. Therefore, our experimental focus shifted toward improving recall while maintaining an acceptable precision level.

We have implemented three different YOLO modes, 5, 10, and 11, with three different YOLO sizes: nano, small, and medium, which resulted, as shown in Fig 3, the best recall 0.902 was achieved by the YOLOv5n, which is better than YOLOv8 model in term of the number of parameters and the inference time as well as in Table I. YOLOv5m achieved a similar recall with 0.9, but due to a large number of 25M parameters, the performance is slower with an inference time of 3.4ms compared to the inference time of YOLOv5n of 0.8. Some noteworthy models were YOLOv10s, with a recall of 0.896, and YOLOv11n, with a recall of 0.89.

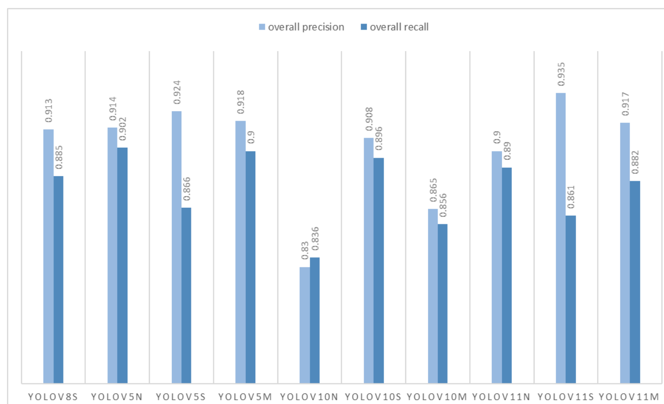

Fig. 3. Overall precision and inference comparison between YOLO models.

TABLE I
Overall Performance Metrics of YOLO Models

Model	Precision	Recall	Params	Inference
YOLOv8s	0.913	0.885	11.1M	1.8ms
YOLOv5n	0.914	0.902	2.5M	0.8ms
YOLOv5s	0.924	0.866	9.1M	1.6ms
YOLOv5m	0.918	0.900	25.0M	3.4ms
YOLOv10n	0.830	0.836	2.7M	1.1ms
YOLOv10s	0.908	0.896	8.0M	2.1ms
YOLOv10m	0.865	0.856	16.5M	4.3ms
YOLOv11n	0.900	0.890	2.6M	0.1ms
YOLOv11s	0.935	0.861	9.4M	1.8ms
YOLOv11m	0.917	0.882	20.0M	4.1ms

As shown in Table II, we presented the per-class recall value to dive more into a detailed analysis. From our observation, we noted that the no-tumor class had the worst recall of 0.689 on the base mode. Although YOLOv10s achieved the highest recall for this class at 0.737, our overall best-performing model, YOLOv5n, also demonstrated a notable improvement, reaching a recall of 0.728. We improved most of the class's recall value, although the class with the best recall value in most models (Pituitary) decreased slightly from 1 to 0.984 on our best model. We noticed that each class has a model that obtained the best recall for it. The model YOLOv5m had the best recall of the Glioma class, whereas YOLOv11n had the best recall value for Meningioma, and, finally, the base model YOLOv8s and YOLOv11n performed the same on Pituitary with 1 recall value.

TABLE II
RECALL PERFORMANCE OF YOLO MODELS ON TUMOR TYPES

Model	Glioma	Meningioma	No Tumor	Pituitary
YOLOv8s	0.931	0.919	0.689	1.000
YOLOv5n	0.952	0.945	0.728	0.984
YOLOv5s	0.912	0.927	0.663	0.959
YOLOv5m	0.966	0.945	0.705	0.982
YOLOv10n	0.871	0.909	0.591	0.973
YOLOv10s	0.952	0.932	0.737	0.964
YOLOv10m	0.935	0.864	0.653	0.973
YOLOv11n	0.935	0.948	0.677	1.000
YOLOv11s	0.927	0.918	0.642	0.955
YOLOv11m	0.927	0.951	0.705	0.945

VI. CONCLUSION

In conclusion, this study presents a comprehensive testing of YOLO-based models for the detection and multi-class classification of brain tumors using MRI scans. Among the evaluated YOLO versions, YOLOv5n gave the best performance in all classes in terms of recall and efficiency, while YOLOv5m, YOLOv11m, YOLOv11n, and YOLO10s achieved the highest per-class recall for Glioma, Meningioma, Pituitary, and No Tumor, respectively. The results indicate the promising abilities of YOLO models in clinical decision-making, particularly in healthcare settings that lack radiological expertise and increasing diagnostic demands. Moreover, our findings show that using real-time object detection models, such as YOLO, can considerably improve early tumor detection. For future work, we will focus on improving recall for the class "no tumor", due to it achieves poor performance across several YOLO models. Moreover, to further enhance the models' performance, we will be training them on more diverse and augmented datasets to improve the models' generalization and robustness. Finally, deploying the optimized model in a real clinical setting through a user-friendly interface will be investigated to determine real-world performance and usability.

VII. AUTHOR CONTRIBUTIONS

J. Alshangiti developed the research idea and contributed to the introduction, results, and discussion sections.

R. Alkhuraiji conducted the gap analysis and prepared the dataset description.

M. Almajed contributed to the development of the methodology section.

R. Alomair implemented the code and contributed to the results and discussion sections.

L. Almutairi contributed to the abstract and conclusion sections.

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